

ST. ANDREW'S JUNIOR COLLEGE



JC2 Preliminary Examination

H1 Chemistry (8872)

22nd September 2011

Paper 1

50 minutes

Additional materials: Data Booklet
Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, class and index number on the Answer Sheet in the spaces provided.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider to be correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

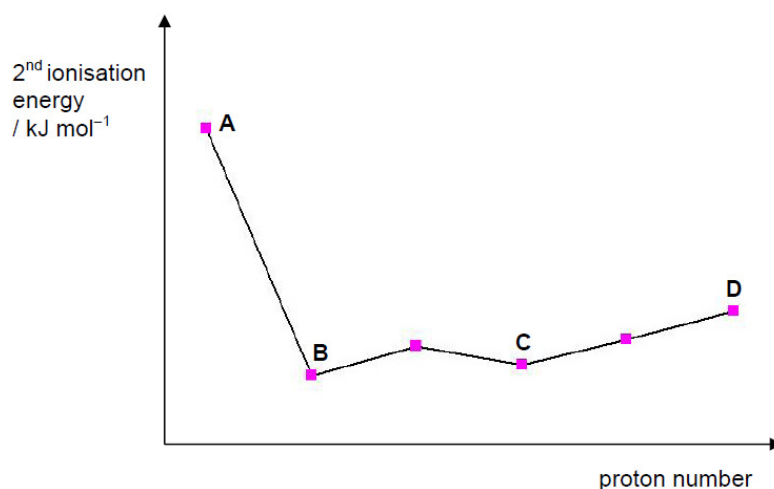
This question paper consists of **13** printed pages including this page.

Section A (25 marks)

For each question there are four possible answers, **A**, **B**, **C** and **D**. Choose the one you consider to be correct.

- 1** Which of the following quantities is equal to the Avogadro constant?
- A** The number of water molecules in 22.4 dm^3 of water at 273 K and 1 atm.
 - B** The number of molecules in 1 dm^3 of oxygen at 273 K and 1 atm.
 - C** The number of atoms in 5.6 dm^3 of ammonia at 298 K and 1 atm.
 - D** The number of atoms in one mole of hydrogen gas at 298 K and 1 atm.
- 2** In an experiment, 10 cm^3 of a hydrocarbon, **X**, was exploded with 100 cm^3 of oxygen present in excess. The volume of gaseous products was 80 cm^3 , but this decreased to 40 cm^3 after treatment with aqueous alkali. All volumes were measured under room conditions. The molecular formula of **X** is
- A** C_2H_4
 - B** C_2H_6
 - C** C_4H_8
 - D** C_4H_{10}
- 3** *Use of the Data Booklet is relevant to this question.*
- An aqueous solution contains 1 mole of $\text{S}_2\text{O}_3^{2-}$ ions and this reduces 4 moles of bromine molecules. What is the sulfur-containing product of this reaction?
- A** H_2S
 - B** $\text{S}_4\text{O}_6^{2-}$
 - C** SO_4^{2-}
 - D** $\text{S}_2\text{O}_8^{2-}$

- 4 The variation in the second ionisation energy of six consecutive elements in the Periodic Table is shown in the following graph.
- Which of these elements is a Group II element?



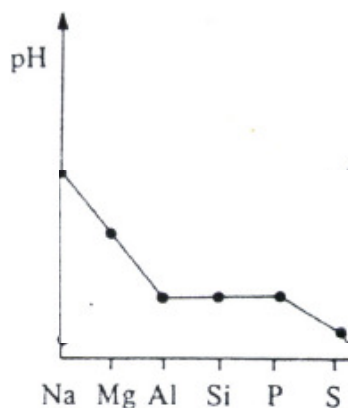
- 5 Which of the following isoelectronic ions, Cl^- , S^{2-} and K^+ , are arranged in the order of increasing size?
- A** K^+ , Cl^- , S^{2-}
- B** K^+ , S^{2-} , Cl^-
- C** S^{2-} , Cl^- , K^+
- D** Cl^- , S^{2-} , K^+

[Turn over]

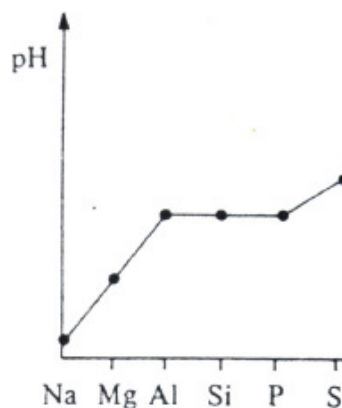
- 6 In separate experiments, 1 mole each of the chlorides of the elements sodium to sulfur is added to water.

Which of the following graphs best represents the variation of pH of the solutions produced?

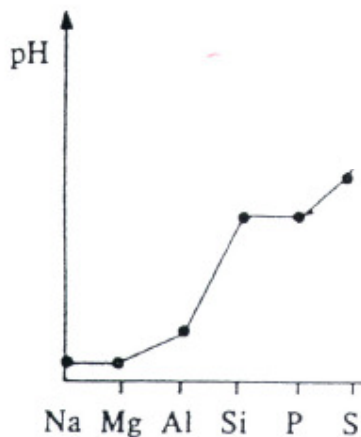
A



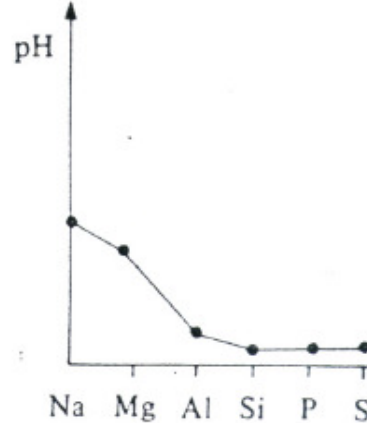
B



C



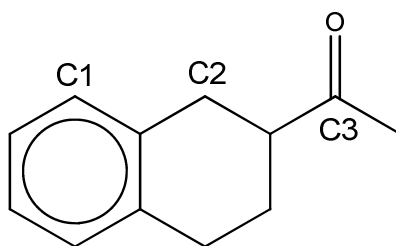
D



- 7 Which of the following does **not** show an overall dipole moment?

- A SF_4
- B IF_5
- C XeF_4
- D ICl

- 8 What are the hybridisation states of the labelled carbon atoms?



- | | C1 | C2 | C3 |
|---|-----------------|-----------------|-----------------|
| A | sp ² | sp | sp ² |
| B | sp | sp ³ | sp ² |
| C | sp ² | sp ³ | sp ² |
| D | sp ² | sp ² | sp |
- 9 Which of the following shows the correct trend in electrical conductivity?
- A Si > Mg > Na > S
B Na > Mg > Si > S
C Si > Na > Mg > S
D Mg > Na > Si > S
- 10 In which of the following sets does the first compound have a larger bond angle than the second compound?
- A H₂S and H₂O
B ClO₃⁻ and BrO₃⁻
C PCl₃ and NCl₃
D BCl₃ and BeCl₂
- 11 The standard enthalpy change of formation of ethane is -84 kJ mol⁻¹.
Given that the standard enthalpy change of combustion for carbon and ethane are -393 kJ mol⁻¹ and -1560 kJ mol⁻¹ respectively, what is the standard enthalpy change of combustion of hydrogen?
- A -286 kJ mol⁻¹
B -417 kJ mol⁻¹
C +230 kJ mol⁻¹
D +361 kJ mol⁻¹

[Turn over]

- 12 A student carried out an experiment to determine the enthalpy change of combustion of octane. The following data was collected.

Mass of water used / g	50.00
Initial temperature of water, T_i / °C	28.0
Final temperature of water, T_f / °C	34.0
Mass of spirit lamp and octane before combustion, m_{initial} / g	44.52
Mass of spirit lamp and octane after combustion, m_{final} / g	40.00

Which of the following expression shows the calculation of the enthalpy change of combustion of octane?

A
$$\Delta H_{\text{combustion}} = - \frac{(44.52 - 40.00) \times 4.18 \times (307 - 301)}{\frac{50}{114}}$$

B
$$\Delta H_{\text{combustion}} = - \frac{44.52 \times 4.18 \times (34 - 28)}{50}$$

C
$$\Delta H_{\text{combustion}} = - \frac{50 \times 4.18 \times (307 - 301)}{\frac{(44.52 - 40.00)}{114}}$$

D
$$\Delta H_{\text{combustion}} = - \frac{50 \times 4.18 \times (34 - 28)}{(44.52 - 40.00)}$$

- 13 At 25°C, sucrose decomposes in acidic solution into glucose and fructose according to the rate equation below:

$$\text{Rate} = k [\text{sucrose}]$$

1.0 mole of sucrose was allowed to decompose in acidic solution. At the end of 10 hours, it was found that only 25% of the original sample was left in the reaction flask.

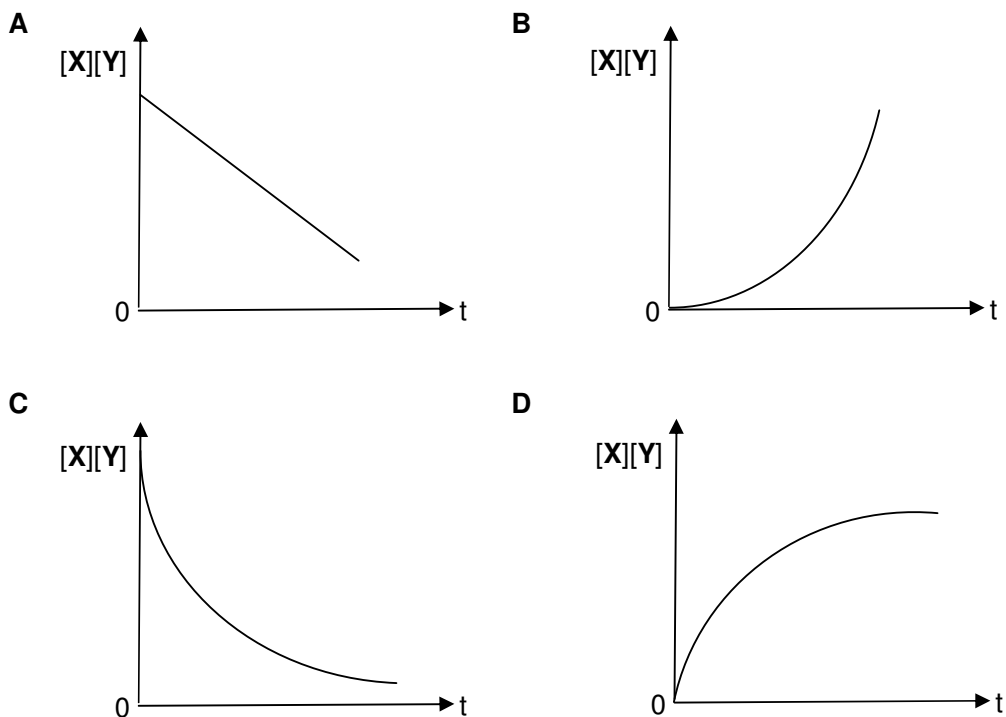
What is the half-life of the decomposition of sucrose in acidic solution?

- A 2.5 hours
- B 5.0 hours
- C 7.5 hours
- D 10.0 hours

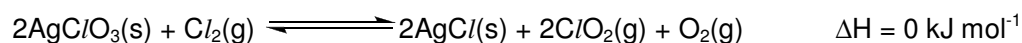
- 14 The product of the concentrations of **X** and **Y**, $[X][Y]$, is plotted against time, t , for the following reaction, where the order of reaction with respect to **X** and **Y** is one respectively.



Which graph would be obtained?



- 15 Silver chlorate(V) reacts with chlorine as follows:

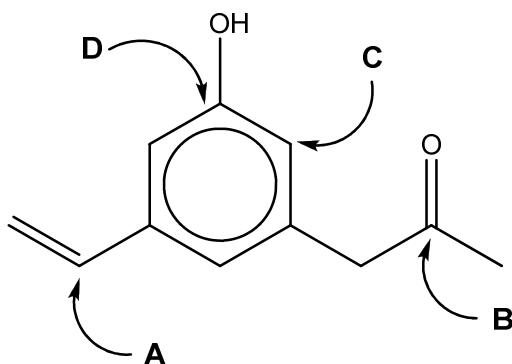


Which of the following change would affect the equilibrium position of the above reaction?

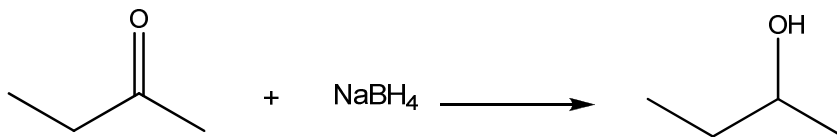
- A** An increase in temperature
- B** The addition of a suitable catalyst
- C** The addition of an inhibitor
- D** An increase in pressure

[Turn over]

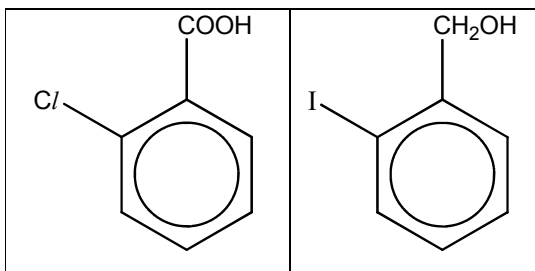
- 16 Which one of the following pairs will **not** give a buffer solution when equal volumes of the two solutions are mixed?
- A $0.2 \text{ mol dm}^{-3} \text{ C}_6\text{H}_5\text{COOH}$ and $0.1 \text{ mol dm}^{-3} \text{ KOH}$
- B $0.1 \text{ mol dm}^{-3} \text{ NH}_3$ and $0.1 \text{ mol dm}^{-3} \text{ NH}_4\text{Cl}$
- C $0.1 \text{ mol dm}^{-3} \text{ NaHCO}_3$ and $0.1 \text{ mol dm}^{-3} \text{ HCl}$
- D $0.2 \text{ mol dm}^{-3} \text{ NaHCO}_3$ and $0.1 \text{ mol dm}^{-3} \text{ NaOH}$
- 17 The acid-base indicator, bromocresol green has a pH range of 3.8 – 5.5. The acidic colour of the indicator is yellow and the alkaline colour is blue.
- Which one of the following statements about the indicator is **true**?
- A In pure water at room temperature, the indicator has a green colour.
- B In a solution containing $8.0 \times 10^{-5} \text{ mol dm}^{-3}$ barium hydroxide, the indicator has a green colour.
- C In a solution containing $1.0 \times 10^{-6} \text{ mol dm}^{-3}$ hydrochloric acid, the indicator has a yellow colour.
- D The indicator is suitable for the titration between hydrochloric acid and ammonia.
- 18 Which of the following carbon positions will be most susceptible to a nucleophile's attack?



- 19 Which of the following correctly describes the reaction below?



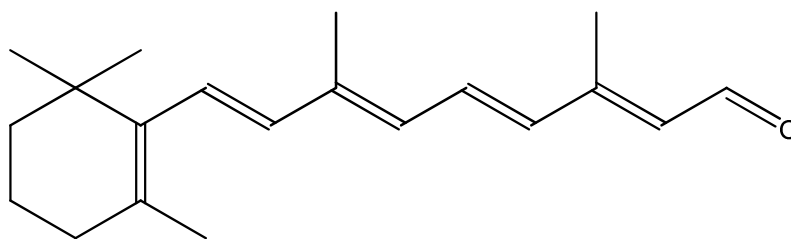
- A Addition
B Oxidation
C Substitution
D Condensation
- 20 Which of the following reagents could distinguish between the two compounds?



- A Sodium metal
B Aqueous sodium carbonate
C Aqueous silver nitrate
D Phosphorus(V) chloride
- 21 How many alcohol structural isomers would $\text{C}_5\text{H}_{12}\text{O}$ have?
- A 3
B 6
C 8
D 9

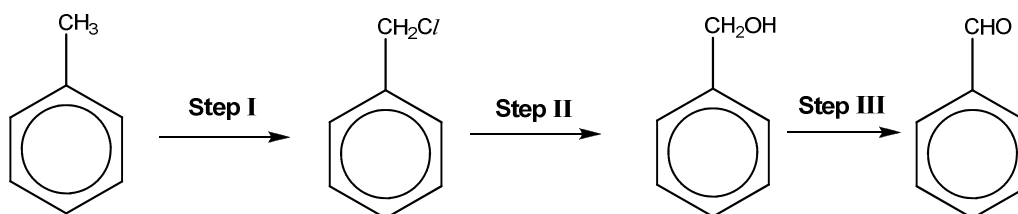
[Turn over]

- 22 β -Apo-15-carotenal is a chemical compound found in fruit extracts. How many stereoisomers does it have?



β -Apo-15-carotenal

- A 4
B 8
C 16
D 64
- 23 Which of the following shows the correct set of reagents and conditions for the synthesis below?



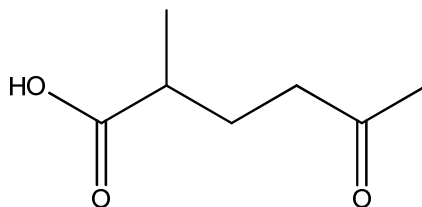
Step I

Step II

Step III

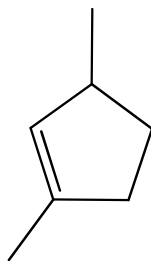
- | | | | |
|----------|---|---------------------------------|---|
| A | $\text{Cl}_2(\text{g})$, FeCl_3 | $\text{NaOH}(\text{aq})$, heat | $\text{H}^+/\text{K}_2\text{Cr}_2\text{O}_7$, heat and distill |
| B | Cl_2 , uv light | $\text{NaOH}(\text{aq})$, heat | $\text{H}^+/\text{K}_2\text{Cr}_2\text{O}_7$, heat and distill |
| C | Cl_2 , uv light | Ethanolic NaOH , heat | H^+/KMnO_4 , heat and distill |
| D | $\text{Cl}_2(\text{g})$, FeCl_3 | Ethanolic NaOH , heat | H^+/KMnO_4 , heat and distill |
- 24 2-chloroethanal was heated with ethanolic KCN and subsequently reacted with lithium aluminium hydride.
- Which of the following compound could be the product?
- A $\text{CH}_2(\text{CN})\text{CH}_2\text{OH}$
B $\text{CH}_2(\text{Cl})\text{CH}(\text{OH})\text{CH}_2\text{NH}_2$
C $\text{CH}_2(\text{Cl})\text{CH}(\text{OH})\text{CN}$
D $\text{CH}_2(\text{NH}_2)\text{CH}_2\text{CH}_2\text{OH}$

- 25 The compound shown below was obtained after **W** was treated with hot acidified potassium manganate(VII).

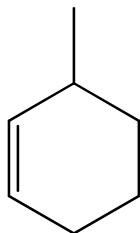


Which of the following could be **W**?

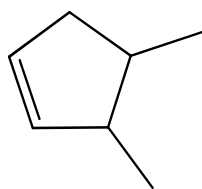
A



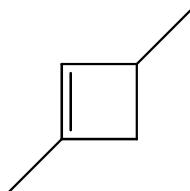
B



C



D



[Turn over]

Section B (5 marks)

For each of the questions in this section one or more of the three numbered statements **1** to **3** may be correct. Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements which you consider to be correct).

The responses **A** to **D** should be selected on the basis of:

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

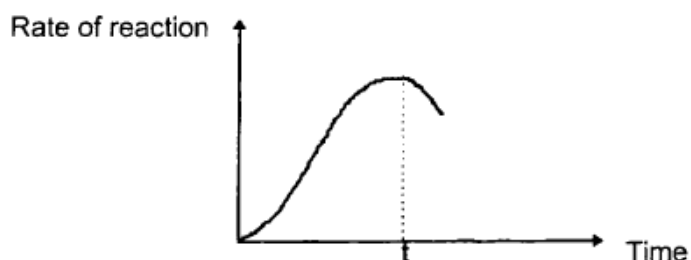
26 Which of the following compounds can form hydrogen bonds with water?

- 1** HCHO
- 2** CH₃COCH₃
- 3** CH₃Cl

27 Which of the following properties **decreases** from NaCl to PCl₃?

- 1** Covalent character
- 2** pH of the resultant solution when mixed with water
- 3** Electrical conductivity of the chloride in liquid state

28 In an experiment, the following graph was obtained when the rate of the reaction was plotted against time.

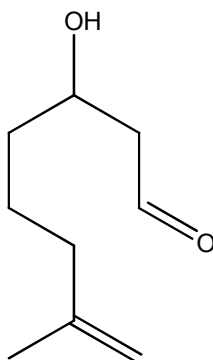


Which of the following conclusions can be made from this graph?

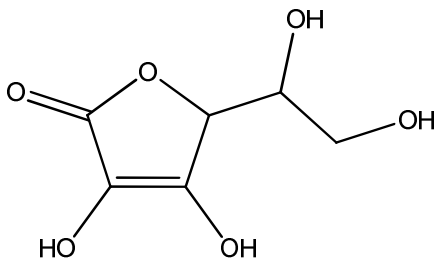
- 1** One of the products catalyses the reaction.
- 2** The reaction is completed at time t.
- 3** The reaction is a first order reaction.

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

- 29 Which of the following reagents would show a positive test when reacted with the compound below?



- 1 Tollen's reagent
 - 2 Aqueous bromine
 - 3 Acidified potassium dichromate(VI)
- 30 Vitamin C has the following structure. Which of the following statements are true for the reactions of vitamin C?



- 1 Vitamin C will react with hot aqueous sodium hydroxide
- 2 Vitamin C will show an orange precipitate with 2, 4 - dinitrophenylhydrazine
- 3 One mole of Vitamin C will react with sodium to produce 4 moles of hydrogen gas.

*****End of paper*****